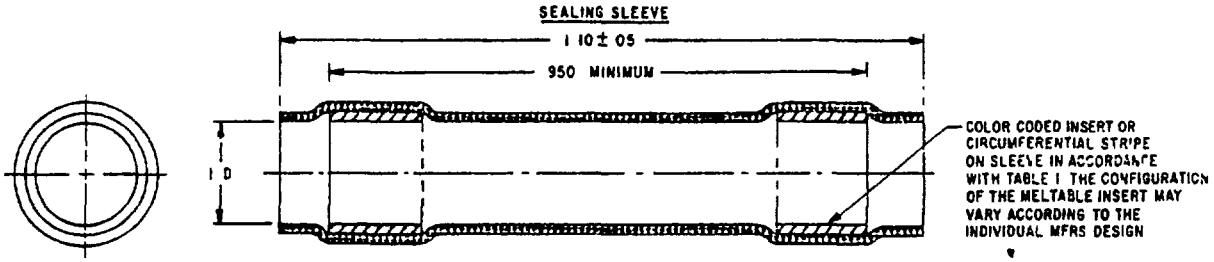
MILITARY SPECIFICATION SHEET

SPLICE ELECTRIC PERMANENT CRIMP STYLE COPPER INSULATED ENVIRONMENT RESISTANT CLASS 1

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the splice described herein shall consist of this document and the latest issue of MIL-S-81824



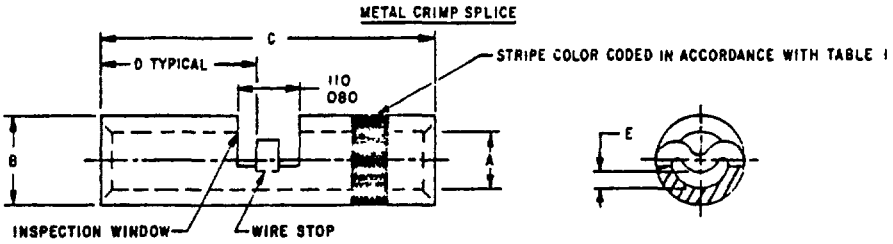


TABLE I CONSTRUCTION DETAILS

PART NUMBER	WIRE RANGE	I D * a (MIN) b (MAX)	A	B	C	D	E MAX	COLOR CODE	WEIGHT LBS/MPC (MAX)
M81824/1-1	26-24-22-20	.085 / .025	.053 / 045	.080 / 075	.510 / 490	.245 / 225	015	RED	1 10
M81824/1-2	20-18-16	.110 / 025	070 / 063	107 / 100	.585 / 565	.280 / 260	020	BLUE	1 65
M81824/1-3	16-14-12	.170 / 025	105 / 095	.155 / 146	.585 / 565	.280 / 260	050	YELLOW	2 75

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
015	38	053	1 35	085	2 16	110	2 79	245	6 22	565	14 35
020	51	063	1 60	095	2 41	147	3 73	260	6 60	585	14 86
025	64	070	1 78	101	2 57	155	3 94	280	7 11		
045	1 14	075	1 91	105	2 67	170	4 32	490	12 45		
050	1 27	080	2 03	107	2 72	225	5 72	510	12 95		

* I D a-"AS RECEIVED"
b-"AFTER UNRESTRICTED RECOVERY THROUGH MELTABLE MATERIAL"

QUALIFICATION — FOR QUALIFICATION TESTING ONLY SPLICE ASSEMBLIES MADE WITH WIRE CONFORMING TO M22759 11 SHALL BE TESTED WITH THE METAL SPLICE THE SEALING AND INSULATING SLEEVE SHALL BE INSTALLED WITH THE APPLICABLE HEAT TOOL FOR CRIMPING TOOLS SEE TABLE II

MATERIAL — COPPER PER WW-T-775 OR ASTMB75-68 OR QQ-C-502 OR QQ C-576
INSULATION SLEEVE HEAT SHRINKABLE TRANSPARENT POLYVINYLIDENE FLUORIDE PER THE PERFORMANCE REQUIREMENTS OF MIL-I-23053 8
SEALING MATERIAL MODIFIED THERMOPLASTIC, SEE PROCUREMENT SPECIFICATION

FINISH — METAL, TIN-PLATED SEE PROCUREMENT SPECIFICATION

DIMENSIONS ARE IN INCHES
METRIC EQUIVALENTS (TO THE NEAREST 01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25 4 MM
UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE 005 (13 MM) FOR THREE PLACE DECIMALS AND 01 (25 MM) FOR TWO PLACE DECIMALS
THE AVERAGE OF DIAMETER "A" AND 'B' MUST BE WITHIN SPECIFICATION LIMITS
'B' DIAMETER SHALL NOT INCLUDE COLOR STRIPE
CONTOUR MAY VARY FROM THAT SHOWN
NO ETCHING OR SOLVENT TREATING OF THE ASSOCIATED WIRE INSULATION SHALL BE PERMITTED

FSC 5940

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:14Z
Check the source to verify that this is the current version before use.

TABLE II CRIMP TOOLS		
SPLICE PART NO	BASIC CRIMP TOOL NO	DIE NO
M81824 1-1 M81824 1-2	M22520 5-01 OR M22520 10-01	M22520 5-103 OR M22520 10-104
M81824, 1-3	M22520 5-01 OR M22520 10-01	M22520 5-102 OR M22520 10-103

Custodians
Navy - AS
Army - EL
Air Force - 85

Review activities
Navy
Army - AV
Air Force - 11, 80
DSA-GS

User activities
Navy - MC, EC
Army - MU
Air Force

Preparing Activity
Navy - AS
(Project No 5940-0772)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:14Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-81824/1 SPLICE, ELECTRIC, PERMANENT, CRIMP STYLE COPPER, INSULATED, ENVIRONMENT RESISTANT, CLASS 1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:14Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department(Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:14Z
Check the source to verify that this is the current version before use.